MySQL extension: User Defined Types

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MySQL User Defined Types: motivation (why ?)

- MySQL:
  - Has builtin SQL types (INTEGER, VARCHAR)
- An application built on MySQL:
  - May use a data type over and over again (for example, IP addresses)
- Problem:
  - “IP Address” is not a builtin data type in MySQL
  - How to represent an IP address in a table ? VARCHAR ? BINARY ?
  - How to perform operations on IP addresses (Is IP X within subnetwork Y) ?
  - Solution must be space efficient (storage), CPU efficient (DML), easy to use (SQL Queries), safe, reusable, …

MySQL User Defined Types: user experience (what ?)

- MySQL server:
  - Introduce user defined types
    - CREATE TYPE IPv6_type AS …
  - Use user defined types in tables
    - COLUMN my_ip IPv6_type
- A component:
  - Provides a supporting implementation for the type (IPv6_type)
  - Provides operations (function IPv6_in_subnet(ip, subnet))
- An application:
  - Declares a type
  - Loads the component providing the type implementation
  - Can use the type as if it was a builtin type.

MySQL User Defined Types: architecture (how ?)

- New SQL construct:
  - CREATE TYPE, DROP TYPE
- New DATA DICTIONARY content:
  - TABLE INFORMATION_SCHEMA.USER_DEFINED_TYPES
- New component services
  - Register a type implementation
  - Register operations implementation
- New builtin types (UUID, implemented as a static component)
- New dynamic types (delivered as loadable components)
- Type ecosystem: third parties can provide their own

MySQL User Defined Types: implementation (details)

- Embedded type:
  - CREATE TYPE IPv6 AS BINARY(16)
    - TYPE IPv6 is what an application manipulates in SQL (logical)
    - Embedded BINARY(16) is what is stored (physical)
- Operations:
  - INSERT into t1 VALUES (“192.168.0.0”) --> to decide: illegal ? Implicit ?
  - INSERT into t1 VALUES (ipv6_from_string(“192.168.0.0”)) --> valid
- INSTALL COMPONENT “file://component_ipv6”
  - Converts between logical and physical representations.
  - Implements all operations (ipv6_from_string, ipv6_to_string, ipv6_in_subnet)

MySQL User Defined Types: code impact

- Some types are static, known at compile time (builtin, MYSQL_TYPE_INTEGER)
- Some types are dynamic, known at runtime (user defined types, IPv6)
- Concept of “type” needs to be abstracted
  - Replace enumerated types by “type descriptors” objects
  - Everywhere a value is used in the server
- Impact on:
  - Parser, optimizer, Data Dictionary, MDL, privileges
  - Prepared statements (text and binary protocol)
  - Stored programs (procedures, functions, triggers, events)
  - Install / Upgrade / Backup / Restore / Clone
  - Open questions: Clients ? Connectors ? Protocol new capabilities ?

MySQL User Defined Types: engineering challenges (design)

- What if the component is not loaded ?
  - Need to have reasonable behavior in degraded cases
    - Backup / Restore
    - Install / Upgrade
    - Replication
    - Cross version compatibility
- Type safety versus implicit type conversions
  - Don’t break existing applications (loose typing)
  - Don’t break the data (strong typing)
- Database migration when altering an existing type (USBN10 to USBN13)
- Many more ...

MySQL User Defined Types: engineering challenges (code)

- Refactoring in 26.x and merge collisions with MySQL 9.7-LTS
- Volume of code affected
- Numbers of code areas affected (every team / code base)

MySQL User Defined Types: proposed development roadmap

- All public in github
  - https://github.com/mysql/mysql-server/issues/674
- Proof Of Concept:
  - https://github.com/mysql/mysql-server/pull/704
- Distributed development & Contributions
  - Collaboration on spec, design, implementation
- Incremental delivery
  - Feature flag at compile time
  - Implement, test, review peice by peice
  - Incremental merges into MySQL 26.x

Thank You

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